

All-Out Organizational Scrum as an Innovation Value Chain

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Abstract

When Scrum extends beyond software development into the organization, Agile techniques, notably Scrum, provide a built-in innovation process. There is more to innovation than coming up with good ideas. Innovating involves capitalizing on well implemented ideas. Implementing good ideas involves risk and consequently should be managed carefully.

Scrum effectively implements Lean methods into organizations by providing simple yet effective tools and techniques. These tools and techniques extend beyond software. Unfortunately, because of Scrum's success and roots in software project management, the language of Scrum is software-centric. This paper describes the Scrum language from a business perspective so when Scrum extends into the organization, innovation in the form of value creation is a natural by-product.

Two companies are in the process of implementing organizational Scrums, one a professional services organization and the other, an Internet-based product company. The description of all-out Scrum in this paper is based on these efforts.

Creating Customer Value

Staying in business in today's global economy requires conducting business in a way customers consider valuable. With so much competition and increased pressure to deliver customer value quickly, the ability to innovate is paramount.

All-out Organizational Scrum provides an innovation value chain by implementing transparency across the organization, managing the idea process, using short Sprint cycles to evaluate the output of the cycles, encouraging frequent releases (diffusion) and removing obstacles in priority order. The obstacles are waste in the organization that are being removed which leans out the organization.

2. Scrum in eighty-seven words

Scrum is an Agile process that allows us to focus on delivering the highest business value in the shortest time. It allows us to rapidly and repeatedly inspect actual working software every two weeks to one month. The business sets the priorities. Our teams self-manage to determine the best way to deliver the highest priority features. Every two weeks to a month anyone can see real working software and decide to release it as is or continue to enhance for another two weeks to a month. [1]

3. Scrum as a Lean implementation

Though the roots of Scrum were partially based on the study of Lean Companies by Takeuchi and Nonaka, Scrum wasn't specifically called "Lean." [2] The connections lie in the study of complex adaptive systems. [3] Many organizations who have modified their software development system based on Scrum consider their work a Lean implementation.

Even in its simplest state, Scrum uses a lean "Pull" technique to smooth the flow through the system and prevent overloading. This pull system exists in the process of Sprint Planning where the team is empowered to select only the amount of work it can reasonably take on. In a Sprint, the team is expected to take on the accountability to do everything it can to complete the Sprint plan on time with high quality.

Scrum also implements a process for eliminating Muda, or waste. [4] A Scrum team is encouraged to identify impediments. When they cannot remove the impediments themselves, the ScrumMaster works to remove them or help the organization understand and eliminate them in priority order. These impediments can be categorized into various types of waste. This continuous pursuit of perfection through removal of waste is called Kaizen [5]. Waste in most organizations consists primarily of: excess inventory,

work-in-progress, delay, multi-tasking and over-production.

The lean systems inherent in Scrum help focus organizations on value. Value in lean terms is defined as, “a capability provided to a customer at the right time at an appropriate price, as defined in each case by the customer.” [6].

4. Scrum as an innovation process

Innovation is the art of developing a new product, service or process based upon a new idea. Innovation is the physical form of the idea—a new product, service or process that you can use to make money. [7] In today’s global economy innovation is the lifeblood of any organization.

Innovation is “a creative idea that is realized.”[8] Thus, only implemented ideas where customers find value in them can be called an innovation. The process of “Customer Value Creation” [9] is defined by SRI International. SRI describes a champion, someone who evangelizes and works the idea into a coherent vision and ultimately (we hope) into an innovation. Another important component of innovation is the continual refinement and synthesis of ideas.

The Post-it Note story provides a good example of innovation. An employee of 3M, Art Fry was a member of his church's choir and was frustrated by the fact that the slips of paper he used to mark hymns repeatedly fell out of his hymnal. What he needed was a sticky bookmark that did not harm the book. He knew of a new adhesive being developed by Spence Silver at 3M and realized that perhaps, just perhaps it could solve his problem. It did but only after five years of additional research and development. The Post-it Note was finally launched in 1980 and was an instant success. There are now more than 1,000 varieties of it on the market and together, they generate hundreds of millions of dollars in annual sales. After a 40-year career at 3M, Fry retired. "If we discover something, we have a chance to stop and look at it. This is very important because lots of things are discovered and passed by because everybody's too busy." [10]

The Innovation Value Chain [11] describes an integrated flow: Idea Generation, Idea Conversion, and Idea Diffusion. “Viewing innovation as an end-to-end process” requires transparency and a highly effective decision process that helps stakeholders at all levels to influence the process without increasing the churn within the organization.

Creating breakthrough technologies creates influences that can set directions for decades or even centuries. Strategies for innovation that are similar to Scrum have been applied throughout history. For

example, Fillippo Brunelleschi (Florence, 1377-1446) became known as the “renewer of Roman masonry work” and is credited with starting the architectural renaissance that lasted hundreds of years. His harmony and economy required him to invent and implement many new technologies and processes to accomplish his great architectural feats of large open space without armature. Brunelleschi employed constant “searching-studies of simple things” to accomplish breakthrough innovations. [12]

Scrum constantly searches to simplify complex “things” by iteratively studying small segments of a “thing” and making it simpler by helping it become well-understood. As understanding increases, we adapt our plans to reflect this new knowledge. This ability to inspect and adapt is the essence of empirical process control methods upon which Scrum is based [13]. Further, Scrum accommodates the innovation process of separating ideas with great potential from those that are challenging and exciting, but lack the ability to become or contribute to great products and services. Scrum accomplishes this selection and filtering process by using frequent, formal reviews where judgment is based upon the potential value available against a standard of shippable quality.

5. Innovation adds risk to organizations

It is important to realize that while the payoffs can be great, being innovative means taking risks. Since more ideas are often filtered than fully realized, uncertainty is a tangible risk. Even if it works, will it be what a customer wants? In today’s date-driven environment unexpected delays have high risk. Unexpected costs may render the potential benefits moot.

Scrum manages risk for innovation. Scrum asks one to prioritize work based on business value and risk so risk is not ignored until it is too late. The standard of shippable quality at each review forces an assessment against this standard so the gaps help identify additional yet-to-be-identified risks.

Scrum helps manage the risk of innovation by providing tangible, high-quality feedback based on the most important information needed to make good decisions. Planning meetings are timeboxed so they do not result in analysis paralysis. The transparency of impediment management and waste removal along with continually evaluating progress against a deployment standard acquires the right kind of information so overall risk is reduced.

6. Comparing organizational Scrum and software development

Whether Scrum is used to ship a piece of software or used for the improvement of an organizational system, constant “searching-studies of simple things” is what Scrum does well. Organizations seek a competitive edge and this edge often involves complex systems and technologies. To prevent descending into an unmanageable state, the organization must constantly achieve simplicity in these systems, and in the processes used to create these systems.

For software, the end of each Sprint (also known as an iteration) results in a potentially shippable increment of software. For other disciplines, this may not have a parallel. This is the key to reducing confusion with the Scrum language for organizational Scrum. “Potentially shippable increments” do not always make sense for all business entities.

7. Removing Scrum language conflicts when Scrum is used as an organizational process

Scrum is extremely simple. Whether Scrum is being used to plan a wedding or build software, the language is similar until we have to answer the question, “What are we producing?” Table 1: Language alternatives for non-software Scrums provides some simple language to construct a lexicon for the output of Scrums.

Table 1: Language alternatives for non-software Scrums

Agile and Scrum Language for Software	Language alternatives for non-software Scrums
Software	• Business initiative • Business process • Product • Value-increasing activity
Potentially Shippable Software	• Potential business initiative rollout • Potentially deployable business process • Potentially releasable product • Potentially implementable business value
Release	• Release • Deployment • Rollout
Definition of Done	• Quality standard • Completeness checklist
Acceptance	• Acceptance criteria

Criteria	• Conditions of satisfaction • Change management plan
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As an example, when we state needs like, “As a frequent business traveler I want a simpler way to capture and submit my expenses quickly so I can save time without having to delegate this to an administrator” our potentially implementable business value is clear. This example assumes that reducing the traveler’s administrative overhead will boost the amount of value-increasing activity that can be created without adding additional overhead elsewhere. In one Sprint, we might have several ideas to accomplish this value-increasing activity. A reasonable “definition of done” could be: high level estimate of cost to implement and maintain the new process, time to file a “typical” expense report, and options available for when and where expense reports could be filed. Each option, or potentially implementable business process, is an idea that could become innovative and remove a lot of waste. Sprint 2 could be comparisons of the best candidates based on costs and benefits. This filtering of options could yield a winner (or no winner-keep the old system) by the end of Sprint 2. Deployment may need a change management plan. If this were discussed at Sprint 2 Planning whether it made sense to include this in the acceptance criteria, a rollout could start right away. Otherwise, another Sprint might be needed to achieve a releasable state and roll this out. Retrospectives would help improve how this kind of business initiative could be improved upon next time.

Like Scrum for software, there needs to be a way to prioritize this work so the organization does not get overloaded. The responsible party for prioritization is the Product Owner. When there are many Scrum teams, a special kind of meeting is needed that matches the heartbeat and rhythm of the Scrum cycles to make sure inspect and adapt processes are working. This meeting is called the Meta-Scrum.

8. Prioritization using the Meta-Scrum meeting

In the Meta-Scrum, consensus regarding the priorities, and changes to priorities, gets confirmed. The Product Backlog contains the priorities. Roadmaps representing desired outcomes are compared to the current plan. In an environment of many Scrum teams, competing priorities emerge. The purpose of the meeting is to resolve these conflicts in this meeting. When this happens, the ability to filter ideas becomes an organizational capability. Waste is removed through teams that are innovating internally

while other teams are able to create new products and services.

Conducting a Meta-Scrum requires a strong initial launch to set expectations. Responsible and accountable persons must attend this meeting and be prepared to make decisions or defer authority to the attendees of the meeting. This meeting should occur every week or two. A person who must miss a meeting for legitimate reasons regains influence the next week when he returns. This rule is critical to the success of the Meta-Scrum. Short meeting intervals also help prevent the need for back-channel or side-channel decisions that are more appropriately contained in the Meta-Scrum. A Big Visible Chart or handouts of the Roadmap should always be available along with prioritized backlogs with release plans. Discussions are centered on changes to the roadmaps and release plans. Obstacles and impediments are raised and action plans captured. Decisions are documented and notes published. It is useful when attendees include team representatives as observers to increase transparency. [14]

9. Organizational Scrum in Professional Services

Scrum is a great simplifier. People often have the energy to start something simple. In our first organization, company A, a software technology professional services company, Scrum with XP has been the official method for software development for four years. The most valuable organizational transfer of learning into other parts of the organization during these years has been Scrum and “Big Visible Charts.” The physical nature of these information radiators generates interest in all parts of the company. It changes the way people converse and consider their work. Discussions ensue and others identify techniques for inspection and adaptation in their own parts of the organization, helping seed change. One goal is to seek and “find small changes that have big improvements.” This helps us choose valuable mechanisms over “process for the sake of process.” An example of this is in recruiting. [15] A need to bridge communication between sales and recruiting caused the creation of a task board that was based on a Scrum task board and a Kanban board. It is physically sized to signal a full bin and has a “board worthy” state each item must pass to be allowed to go on the board. Organizational reports contain metrics on board worthy and non-board worthy requisitions separately. Daily meetings were abandoned for spontaneous pull-based

discussions. They seem to take place in front of the board often.

The company has a daily meeting at the end of each day where the leadership meets to address impediments identified by anyone in the company. The meeting is called “The Daily Triage.” It is an open meeting so anyone can come meet and discuss issues with senior management. Rather than force tool usage, anyone who has an impediment can log it or join the meeting to discuss it. If the impediment still exists by the end of the meeting it gets logged into the system for further handling until resolution.

Most corporate initiatives are captured in backlogs and prioritized. The biggest issue with this company is the plethora of ideas. Scrum helps lean this out by prioritizing work through this enterprise product backlog [16]. It helps the organization to learn the discipline to stop doing things that are ultimately distracting so the really good ideas can become innovations.

Table 2: The Scrum language used in Company A

The Daily Triage	Daily Impediment management of waste and risk
Product Backlog	The product backlog is used for prioritization of every team.
Daily Scrum	All software teams and most departments have these. The President likes having Daily Scrums with leadership. Some call this a stand up and others call it a daily meeting. The organization is good about keeping these to 15 minutes.
Big Visible Charts	These are on almost every wall in the company. They include task boards, burndown charts, impediment lists and team working agreements.
Task Boards	Recruiting has a Task Board that incorporates some Kanban elements Most Scrum teams in development and marketing use Task Boards to manage Sprints though some prefer to use software tools

Sprint Planning, Reviews and Retrospectives are held with the many software teams but these are less frequent with non-software, non-project based teams.

It is worth noting several “lean for office” events yielded plans for improving value streams but no direct value has yet to be implemented from this effort. Scrum naturally adopted lean principles with greater impact. It is important to understand lean concepts and

value propositions in order to help optimize Scrum implementations.

10. Organizational Scrum in a Product Company

Company B is an Internet-based product company. They had a division who had been very successful delivering software using Scrum for two years. The executive team wanted to improve its organizational focus and decided to form themselves into a Scrum team named “Team Velocity (v-Team for short).” The objectives were: Gain visibility into the organization, connect people to the business, enable organizational alignment, get employee commitment, hold accountable, increase speed to market, focus growth, increase transparency, simplify and guide opportunities.

This company uses: Agile chartering, five levels of planning [17], daily meetings. Sprint planning is fairly short. Product backlog is prioritized and estimated upon entry on a two dimensional backlog representing size and priority. Sprint reviews to the business are monthly via an all-company meeting

Executives attend every Sprint Review every two weeks. Mid-Sprint, each Product Owner meets with the executive team so the Product Owner can make sure there is stakeholder consensus. Each Product owner has an executive sponsor that is part of “Team Velocity.”

There are significant numbers of executive discussions and decisions that would be inappropriate to share with the entire organization. Except for these, the organization executes using Scrum.

Table 3: The Scrum language used in Company B

Daily v-Team meeting	Team Velocity meets daily
Agile Charter	Include Jim Highsmith’s Tradeoff Matrix. Jointly created with input from executives, Product Owner and team.
Product Roadmap	Provides high level view of the targeted releases.
Product Backlog	The product backlog is used for prioritization of every team.
Daily Scrum	All software, operations teams and departments have these.
Big Visible Charts	These are on almost every wall in every conference room, most of which have become Scrum team rooms. Overlooking the lake, v-Team Velocity looks through its task board and product backlog.

Task Boards	Scrum teams in use Task Boards to manage Sprints though some prefer to use software tools.
Mid Sprint Product Owner Reviews	A great time to ensure there is still consensus on the plans. This has migrated from a release plan review to a roadmap review to keep the detail more appropriate.

For the v-Team, the output of their first Sprint launched five Scrum teams. After four Sprints, the v-Team had a multi-Sprint retrospective. They found the optimization opportunities in their company provide for true servant leadership at the highest levels of the organization. Transparency in the v-Team helps them focus and they find the mutual accountability a healthy forcing function. As one of the executives said, “We absolutely made the right decision to run our company this way.”

11. Conclusion

Scrum is a simple framework that provides the transparency to take advantage of the lean principles of “pull systems” and removing waste. When applied to the organization and not just software development departments, an improved innovation value chain emerges. Innovation not only applies to parts of the organization it entirety. Companies A and B are finding increased value by extending Scrum beyond the development teams

12. References

- [1] <http://groups.yahoo.com/group/scrumdevelopment/message/17284>
- [2] Takeuchi and Nonaka, “The New, New Product Development Game, Harvard Business Review, 1986
- [3] J. Sutherland, “Is it Lean or is it Scrum?”, <http://jeffsutherland.com/scrum/2007/11/is-it-scrum-or-lean.html>, 2007
- [4] Ohno, T., Toyota Production System: Beyond Large Scale Production, Productivity Press, New York, NY, 1988
- [5] Liker, J., The Toyota Way, McGraw-Hill, New York, NY, 2004
- [6] Womack, J.P. and Jones, T.J., Lean Thinking, Free Press Publications, New York, 1996
- [7] Gorman, T., Innovation, F+W Publications, Avon, MA, 2007.

- [8] <http://en.wikipedia.org/wiki/Innovation>.
- [9] Carlson C. and Wilmot W., *Innovation: The Five Disciplines for Creating What Customers Want*, Crown Publishing Group, New York, NY, 2006.
- [10] M.T. Hansen and J. Birkinshaw, "The Innovation Value Chain", *Harvard Business Review*, June 2007.
- [11] Morris, R., "From 'acorns' of ideas to 'oak trees' of business success", <http://www.amazon.com/review/R1V6GI73D4EDX>, 2008
- [12] Prager, F. D. and Scalglia, G. *Brunelleschi: Studies of His Technologies and Inventions*, MIT Press, Cambridge, MA, 1970.
- [13] Schwaber, K. and Beedle, M., *Agile Software Development with Scrum*, Prentice Hall, New Jersey, 2002.
- [14] B. Barton, "Establishing and Maintaining Top to Bottom Transparency Using the Meta-Scrum", <http://www.agilejournal.com/content/view/629/111/>, 2007.
- [15] D. Anderson, "Agile Directions", <http://www.infoq.com/presentations/Agile-Directions-David-Anderson>, video, min 4:00, 2008
- [16] Schwaber, K., *Enterprise Scrum*, Microsoft Press, Redmond, WA, 2007.
- [17] H. Smits, "Scaling Agile Processes: Five Levels of Planning", <http://www.agilejournal.com/articles/articles/scaling-agile-processes:-five-levels-ofplanning.html>,